

Product Manager Assignment

E-commerce System MVP Design & Basic Technical Understanding

Prepared answer for the assignment. Scope: MVP for a basic e-commerce system covering customer transaction flow and internal operations management.

1. User Roles and Core Scenarios

Role	Core Scenarios
Customer / Buyer	Browse products, search/filter products, view product details, select SKU, add to cart or buy now, submit order, pay, track shipment, confirm receipt, view order history, cancel unpaid orders, request refund when eligible.
Operations Staff / Merchant Staff	Create and edit products, manage SKU prices and inventory, publish or take products off sale, review paid orders, arrange shipment, update tracking information, handle cancellations/refunds and abnormal orders.
Platform Admin	Manage staff accounts and permissions, configure platform-level settings, view operational data, audit abnormal orders, review logs, and support dispute/risk handling.

2. MVP Feature List and Priority

Module	Feature	Priority	Reason
Customer Side	Login / Registration	P1	Checkout can support guest mode in the first version, but login improves order history and customer identification.
Customer Side	Product List	P0	Customers need an entry point to browse available products.
Customer Side	Product Detail Page	P0	Customers must see product information, price, SKU, inventory, and purchase options before ordering.
Customer Side	Product Search / Category Filter	P1	Improves discovery efficiency, but a simple product list can support the first transaction flow.
Customer Side	Shopping Cart	P0	Required for add-to-cart flow and multi-item checkout.
Customer Side	Buy Now	P1	Improves purchase speed but is not mandatory if cart checkout works.
Customer Side	Submit Order Page	P0	Required to confirm SKU, quantity, delivery address, total price, and submit the order.
Customer Side	Payment	P0	Required to complete the transaction.
Customer Side	Order List / Order Detail	P0	Customers need to see payment status, shipment status, and order details.
Customer Side	Cancel Order	P1	Useful for unpaid orders and customer service efficiency.
Customer Side	Refund Request	P2	Can be added after the first paid-order flow is stable,

			depending on business policy.
Admin / Operations Side	Product Management	P0	Products must be created and maintained before they can be sold.
Admin / Operations Side	SKU / Inventory Management	P0	Inventory and SKU price control are required to prevent overselling.
Admin / Operations Side	Order Management	P0	Operations staff must review paid orders and process fulfillment.
Admin / Operations Side	Shipment Management	P0	Required to ship paid orders and update tracking information.
Admin / Operations Side	User Management	P1	Needed for permission control when more staff use the system.
Admin / Operations Side	Data Dashboard	P2	Useful for business monitoring, but not required for the first transaction flow.
Platform	Basic Permission Control	P1	Reduces operational risk by limiting access to sensitive functions.
Platform	Operation Logs	P2	Helpful for audits and troubleshooting, but can be expanded later.

3. Core Process Design

3.1 Customer Order and Payment Process

Stage	Normal Flow	Status / Data Change
1. Select product	Customer opens product detail page, selects SKU and quantity, then clicks Add to Cart or Buy Now.	No order status yet. Frontend displays selected SKU price and stock.
2. Submit order	Customer confirms item, quantity, address, freight, and total amount. Backend re-checks product status, SKU stock, price, and cart validity.	Order is created as PENDING_PAYMENT. Stock may be locked depending on inventory design.
3. Pay	Customer chooses payment channel and completes payment within the payment time limit.	Payment success changes order to PAID. Payment transaction ID and paid time are recorded.
4. Fulfillment	Operations staff views paid order and arranges shipment.	Order remains PAID until shipment is completed.
5. Complete	Customer receives goods and confirms receipt, or system auto-completes after a configured period.	Order changes to COMPLETED.

Exception Scenario	Handling Rule	Status / Result
Insufficient inventory	Backend must use actual SKU inventory as final source of truth. If stock is insufficient, block order submission and show error.	No order is created, or order creation fails.
Product taken off sale but still in cart	At checkout, backend re-checks product status. The item is marked invalid and cannot be purchased.	Cart item becomes invalid; no order for that item.
Payment failure	Keep the order unpaid and allow user to retry before timeout. Show payment failure reason when available.	Order remains PENDING_PAYMENT.
Payment timeout	If user does not pay before the limit, system cancels the order and releases locked stock if stock was locked.	Order changes from PENDING_PAYMENT to CANCELLED.

Price changed after item added to cart	At checkout, show latest price and ask user to confirm. Backend uses latest valid price when creating order.	Order total recalculated before submission.
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3.2 Product Creation and Publishing Process

Stage	Normal Flow	Status / Data Change
1. Create draft	Operations staff enters product name, category, images, description, SKU attributes, price, and stock.	Product status is DRAFT.
2. Validate information	System checks required fields, price format, SKU completeness, stock values, image availability, and category validity.	Invalid fields are highlighted; product cannot be published until fixed.
3. Save product	Staff saves the product as a draft or submits for publishing.	Product data is stored in product database.
4. Publish	Staff clicks Publish after validation passes.	Product status changes to ON_SALE. Product appears on product list and can be purchased.
5. Edit after publish	Staff may edit price, stock, images, or description. Critical changes should be logged.	Updated data is visible to users after save/cache refresh.

Exception Scenario	Handling Rule	Status / Result
Missing required field	System blocks save/publish and shows exact missing field.	Stay in DRAFT.
SKU price or stock invalid	Price must be greater than 0. Stock cannot be negative. SKU data must be complete.	Publish blocked.
Inventory set to 0	Product can remain ON_SALE but purchase button should show Sold Out, or staff can set it OFF_SALE depending on business policy.	ON_SALE but unavailable, or OFF_SALE.
Product taken off sale	Staff clicks Take Off Sale. The product remains in database but is hidden from list and cannot be purchased.	Status changes to OFF_SALE.
Published product in user carts	Cart displays the product as invalid if status becomes OFF_SALE or stock becomes 0.	Customer must remove item or choose another SKU.

4. Simplified PRD - Product Detail Page

Item	Content
Feature Background	The Product Detail Page is the main page where customers understand a product and make a purchase decision. It must clearly display product information, SKU options, price, stock, and purchase actions.
User Goal	As a customer, I want to view detailed product information, choose the correct SKU, confirm price and stock, and add the product to cart or buy it directly.
Page Fields	Product image gallery; product name; sale status; current price; original price; sales count; SKU selector; selected SKU price; selected SKU stock; quantity selector; product description; shipping information; Add to Cart button; Buy Now button; Sold Out / Off Sale state; error messages.
Business Rules	Only ON_SALE products can be purchased. Price should be displayed in the user-facing currency format. SKU selection is required before purchase when multiple SKUs exist. Quantity cannot exceed selected SKU stock. If stock is 0, purchase actions are disabled. If status is OFF_SALE, the product can be shown as unavailable but cannot be purchased. Backend validation is final when submitting cart or order.

Error / Exception Handling	If no SKU is selected: "Please select a product option." If quantity exceeds stock: "Purchase quantity exceeds available inventory." If product is off sale: "This product is currently unavailable." If stock is 0: "Sold out." If API fails: "Product information failed to load. Please try again."
Acceptance Criteria	1) Product image, name, price, stock, SKU and sales count display correctly from API. 2) Selecting a SKU updates price and stock. 3) Purchase buttons are disabled for OFF_SALE or zero-stock products. 4) Quantity validation works before Add to Cart or Buy Now. 5) Backend errors are displayed clearly without creating invalid orders.

5. API Understanding Answers

- Information that can be displayed: product ID, product name, sale status, current price, original price, total stock, sales count, product images, SKU names, SKU IDs, SKU prices, and SKU stock.
- price: 19900 should be displayed as 199.00 in the user-facing currency, assuming the API returns price in cents/minor currency units. For example, RMB 199.00 or \$199.00 depending on the platform currency.
- If status = OFF_SALE, the page should show the product as unavailable/off sale, disable Add to Cart and Buy Now, and prevent checkout. The product may still be visible through old links but should not be purchasable.
- If total product stock is 0, the purchase button should be disabled and displayed as Sold Out / Out of Stock.
- If the user selects the Black SKU, the displayed price should change to 209.00 and inventory should change to 15, based on skuld 20002.
- Missing fields: product description, because users need detailed product information; category ID/name, because product listing and filtering need category data; shipping fee or delivery rules, because checkout needs logistics cost; product specifications/attributes, because customers need structured details; created/updated time or version, because operations and cache consistency need data freshness control.

6. Code Logic Reading Answers

- This function determines whether a user is allowed to submit an order for a selected product/SKU and quantity.
- The user cannot submit the order if the product is not ON_SALE, if SKU stock is 0 or below, or if the purchase quantity is greater than available SKU stock.
- If productStatus = 'ON_SALE', skuStock = 3, and buyCount = 5, the result should be: canSubmit = false, reason = 'Purchase quantity exceeds available inventory'.
- Additional checks should include login status, selected SKU validity, delivery address validity, purchase quantity minimum and maximum, purchase limit per user, product price consistency, product/cart validity, payment method availability, delivery area support, risk control, and whether the order amount is greater than 0.

7. Order Status Understanding Answers

- Initial status after customer submits an order: PENDING_PAYMENT.
- After payment is successful: PAID.
- After merchant ships the order: SHIPPED.
- If the customer does not pay within the payment time limit: CANCELLED.
- Customer refund request should usually be allowed after payment, mainly in PAID, SHIPPED, and sometimes COMPLETED depending on refund policy. PENDING_PAYMENT should use cancel order, not refund. CANCELLED and REFUNDED should not allow another refund request.
- Order status transition flow: PENDING_PAYMENT -> PAID -> SHIPPED -> COMPLETED. Exception flow: PENDING_PAYMENT -> CANCELLED when unpaid timeout or user cancels before payment. Refund flow: PAID / SHIPPED / COMPLETED -> REFUNDING -> REFUNDED. If refund is rejected, the order should return to the previous valid status such as PAID, SHIPPED, or COMPLETED.

Status Flow Diagram:

PENDING_PAYMENT -> PAID -> SHIPPED -> COMPLETED
PENDING_PAYMENT -> CANCELLED
PAID / SHIPPED / COMPLETED -> REFUNDING -> REFUNDED

8. Bug / Exception Analysis Answers

Issue: customer sees inventory = 1 on the product detail page, but order submission returns "Insufficient inventory."

Possible Cause	Explanation	Investigation Method
Cached product detail inventory	The detail page may show old stock from CDN, browser cache, or application cache.	Check cache TTL, compare product detail API response with database stock, and confirm whether frontend used cached data.
Another customer bought the last item	Inventory changed between product viewing and order submission.	Check order creation logs and inventory deduction logs for the same SKU during that time window.
SKU stock and total stock inconsistent	Product total stock may show 1, but selected SKU stock may be 0.	Compare product-level stock and selected SKU stock in database/API response.
Cart item became invalid	Product or SKU may have been taken off sale after user added it to cart.	Check product status, SKU status, and cart validation result at order submission.
Backend re-check blocked the order	Backend uses real-time stock validation and is the final source of truth.	Review order submission API logs, request parameters, and backend validation result.
Frontend and backend out of sync	Frontend displayed one value while backend inventory service had another value.	Check API versions, inventory service response, database replication delay, and message queue delay if stock is updated asynchronously.

Recommended handling: keep backend validation as the final source of truth; refresh stock before checkout; show a clear message such as "Only 0 items left, please choose another SKU"; update cart item validity immediately; and monitor inventory cache inconsistency rate.

9. Data Metrics Design

Metric	Meaning	Why It Matters
Product Detail Page Views	Number of visits to product detail pages.	Measures product exposure and traffic quality.
Add-to-Cart Rate	Users who add to cart / users who view product details.	Measures product attractiveness and whether product information is convincing.
Order Submission Rate	Users who submit orders / users who view product details or carts.	Measures checkout intent and checkout usability.
Payment Success Rate	Paid orders / submitted orders.	Shows whether payment flow works smoothly and whether failures are high.
Order Conversion Rate	Paid orders / product detail page visitors.	Measures total funnel performance from browsing to paid order.
GMV	Total value of paid orders.	Measures sales scale and business growth.
Inventory Out-of-Stock Rate	Products or SKUs with stock = 0 / total active SKUs.	Helps operations manage replenishment and avoid lost sales.
Order Fulfillment Time	Average time from payment to shipment.	Measures operational efficiency and customer experience.
Refund Rate	Refunded orders / paid orders.	Indicates product quality, expectation mismatch, or after-sales issues.